Factors Associated with Hospital Admission After Emergency Department Injury Visits: A Nationally Representative Analysis Using NEISS 2024 Data.

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Group 5

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1. Introduction

Injuries are a major cause of emergency department visits, with outcomes ranging from same-day discharge to hospitalization or death. Identifying factors associated with more severe outcomes is important for emergency care planning and injury prevention.

Many emergency department injury patients are treated and released, while others are admitted, transferred, observed, or die, and it is not always clear which patients and injury factors are most strongly linked to hospital admission and severe outcomes.

This project analyzes nationally representative 2024 emergency department injury data from the National Electronic Injury Surveillance System (NEISS) to examine how demographic, injury-related, and incident characteristics are associated with patient disposition. Core course concepts, including data cleaning, Exploratory data analysis, statistical inference, and predictive modeling, are applied using a real-world healthcare dataset.

2. Related Work

Prior research has identified several key factors associated with hospital admission in emergency department (ED) settings. Demographic characteristics, particularly age, have consistently been shown to be strong predictors of hospitalization risk, with older patients more likely to require admission due to higher injury severity and comorbidities (Feretzakis et al., 2021).

Injury-related factors, such as diagnosis type and affected body part, have also been widely studied as indicators of severity and clinical decision-making. For example, fractures and internal injuries are more likely to result in admission compared to minor injuries. Clinical assessment tools applied at triage have also been shown to support early identification of high-risk patients in the ED (Anderson et al., 2013).

Statistical methods such as hypothesis testing and logistic regression are commonly used to analyze associations between predictors and outcomes in healthcare data. More recently, machine learning approaches have been applied to improve predictive performance in ED triage and admission prediction tasks.

However, many studies focus either on statistical inference or predictive modeling, but rarely integrate both approaches with real-world deployment. This project addresses this gap by combining statistical analysis, machine learning modeling, and an interactive application to support admission risk prediction.

3. Methods

3.1 Data Source

The primary dataset used in this project is the 2024 NEISS public-use injury dataset provided by the U.S. Consumer Product Safety Commission. The dataset contains information on patient demographics, injury diagnoses, injured body parts, incident locations, disposition outcomes, and survey design variables, including strata, primary sampling units, and statistical weights, which indicate that the data are nationally representative.

3.2 Data Preprocessing

The NEISS 2024 dataset was cleaned and transformed to ensure consistency, accuracy, and analytical usability. Column names were standardized, and placeholder values (e.g., 0 indicating “not recorded”) were recoded as missing to avoid misleading interpretations.

Categorical variables (e.g., sex, race, injury characteristics, and disposition) were mapped to descriptive labels and converted to categorical types to improve interpretability. Age was standardized into a continuous variable, including conversion from month-based values.

Missing data were handled using a structured approach based on variable importance. Observations with missing values in key variables (e.g., sex, age, disposition) were removed due to minimal impact (<1%), while missing values in selected categorical variables were retained and labeled as “unknown” to preserve information. Variables with high missingness (>50%) were excluded to reduce potential bias.

A refined analytical dataset was created by selecting relevant variables aligned with the research objective. Additional features were engineered, including temporal variables (month, weekday), age groups, and a binary admission indicator representing injury severity.

These preprocessing steps ensured that the dataset was both analytically robust and suitable for downstream statistical and machine learning methods.

3.3 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to understand variable distributions and identify patterns associated with hospital admission.

Numerical variables, such as age (age_year), were summarized using descriptive statistics, while categorical variables were analyzed using frequency distributions to assess category prevalence and imbalance. Temporal trends were examined by aggregating injury counts across weekdays and over time.

Associations between explanatory variables and hospital admission were explored using group-wise admission rates and cross-tabulations. The variability of admission rates within each variable was used as an indicator of potential importance.

Interactions between variables were further examined by analyzing subgroup admission rates across combined factors (e.g., age group and fire involvement), with results visualized using heatmaps.

Insights from EDA informed variable selection and guided subsequent statistical and predictive modeling.

3.4 Statistical Analysis

Following EDA, statistical methods were applied to formally evaluate the relationships between predictors and hospital admission.

The analysis progressed from simple comparisons to multivariable modeling. A two-sample Welch's t-test was used to assess differences in mean age between admitted and non-admitted patients. Prior to testing, normality was assessed using histograms. Both groups were found to be non-normal (bimodal for admitted, right skewed for non-admitted); however, the Central Limit Theorem justifies proceeding given the large sample sizes (admitted: n = 42,212; not admitted: n = 319,073). Chi-square tests were conducted to evaluate associations between categorical variables and admission. Expected cell counts were verified for all variables — the body_part variable failed the minimum expected count condition (min = 2.45 < 5) and was excluded from interpretation. All other variables met the required conditions. Logistic regression was then applied to assess the combined effects of multiple predictors while controlling for confounding factors. Hospital admission was treated as a binary outcome variable (1 = admitted, 0 = not admitted), with statistical significance evaluated at the 0.05 level.

Given the large sample size, emphasis was placed not only on statistical significance but also on effect size and practical relevance to avoid overinterpretation of trivial differences.

3.5 Machine Learning Modeling

A modeling approach was applied to improve predictive performance by addressing class imbalance and enhancing feature representation.

The dataset was partitioned into training (70%), validation (15%), and test (15%) sets using stratified sampling to preserve the class distribution of the admission outcome. The training set was used for model fitting, the validation set for model comparison and tuning, and the test set for final evaluation.

A baseline logistic regression model was constructed using selected demographic and injury-related variables. Categorical features were encoded using one-hot encoding with a consistent feature structure across datasets. Model performance was evaluated on the validation set using classification metrics and ROC-AUC.

To address the imbalance in the admission outcome, class weighting was applied within the logistic regression model. This approach adjusted the loss function to increase the penalty for misclassifying minority class observations.

Additional features were introduced to capture contextual and interaction effects. These included temporal indicators (e.g., weekend), binary flags for alcohol and drug involvement, a text-derived feature (narrative length), and an interaction term combining age group and diagnosis. The expanded feature set was encoded using one-hot encoding, ensuring alignment between training and validation datasets.

Model performance was compared across iterations using validation metrics, with emphasis on recall for admitted cases and ROC-AUC. The best-performing model was selected based on its ability to balance discrimination and sensitivity to the minority class.

The selected model was evaluated on the held-out test set to assess generalization performance. The test set was used only once to avoid information leakage.

The final model and corresponding feature structure were saved for reproducibility and future application.

3.6 Deployment

The final model was deployed as an interactive web application using Streamlit. This deployment enables users to input injury-related information and obtain real-time predictions of hospital admission risk.

By translating the model into an accessible interface, the project extends beyond analysis to provide a practical decision-support tool for potential real-world use.

4. Results

4.1. Exploratory Data Analysis

This exploratory analysis summarizes injury patterns and factors associated with hospital admission in the 2024 dataset.

Injuries were relatively stable across weekdays and showed a seasonal pattern, rising from April, peaking in September, and declining toward year-end. The age distribution was right-skewed, with most injuries affecting younger individuals but extending into older age groups.

Admission rates varied substantially across injury characteristics. Higher rates were observed in cases involving large body areas, submersion, fire-related events, and older adults (65+), while younger age groups and recreational settings showed consistently lower admission rates. Differences by sex, race, and ethnicity were comparatively small.

Multivariate descriptive analysis further indicated that fire involvement amplified admission risk across age groups, particularly among elderly patients. Elevated admission rates for elderly individuals were observed across several locations, including mobile/manufactured homes and street or highway settings. Some extreme subgroup rates likely reflect small sample sizes and should be interpreted cautiously.

Overall, injury severity—proxied by hospital admission—appears primarily driven by injury characteristics and age, with other demographic factors playing a smaller role. These findings guide variable selection for subsequent statistical modeling.

4.2 Statistical Findings

Across all statistical methods, consistent patterns emerged regarding the factors associated with hospital admission.

Age was identified as the most influential factor. Both hypothesis testing and logistic regression showed that older patients, particularly those aged 65 and above, have substantially higher admission risk.

Injury characteristics, particularly diagnosis type, demonstrated meaningful associations with admission, likely reflecting underlying injury severity. Note that body_part was excluded from chi-square interpretation as its minimum expected cell count condition was not satisfied. However, EDA patterns suggest injury location on the body remains a clinically relevant factor.

Contextual variables, such as fire involvement and location, showed statistically significant relationships with admission; however, their effect sizes were comparatively smaller, indicating a more limited practical impact when considered independently.

Importantly, due to the large sample size, many variables were found to be statistically significant. Therefore, interpretation focused on effect size and real-world relevance rather than p-values alone.

Overall, the results suggest that hospital admission is primarily driven by **patient age and injury severity**, while other factors provide additional but secondary explanatory value.

4.3 Machine Learning Performance

Model	Recall (Admitted)	ROC-AUC
Baseline	0.24	0.85
Class Weighted	0.79	0.85
Enhanced Features	0.81	0.86

The modeling process demonstrates a clear trade-off between predictive performance and practical utility.

The baseline model achieved strong overall accuracy and ROC-AUC, but performed poorly in identifying admitted patients, with very low recall. This highlights the limitation of standard models under class imbalance, where accuracy can be misleading.

By introducing class weighting, the model significantly improved recall for admitted cases, increasing from 0.24 to 0.79. This confirms that class imbalance was a key issue affecting model performance.

Further improvements were achieved through feature engineering. By incorporating contextual variables (e.g., weekend, alcohol, drug involvement, narrative length) and interaction effects, the final model reached a recall of 0.81 and an ROC-AUC of 0.857 on the test set.

Although precision remains relatively low, the model prioritizes recall, which aligns with the healthcare objective of minimizing missed high-risk patients.

Overall, the results show that:

- Addressing class imbalance is critical for improving model usefulness
- Feature engineering adds incremental but meaningful gains
- There is an inherent trade-off between recall and precision

The final model provides a practical and interpretable approach for early identification of high-risk patients, while also highlighting areas for further improvement. This trade-off highlights the importance of aligning model performance with practical objectives, particularly in healthcare settings where failing to identify high-risk patients may have serious consequences.

4.4 ED Admission Risk Predictor

To demonstrate practical applicability, the final model was deployed as an interactive web application using Streamlit. The application allows users to input injury characteristics and receive real-time predictions of hospital admission risk.

The deployed application is available at:

https://neiss2024injuryanalysis-ed-admission-risk-predictor.streamlit.app/

5. Discussion

This project aimed to identify key factors associated with hospital admission following injury-related emergency department visits and to develop an interpretable predictive model.

Overall, the project successfully achieved its primary objectives. Key variables were identified and refined from the original dataset, focusing on demographic and injury-related characteristics. Exploratory data analysis revealed clear patterns, particularly the strong relationship between age and admission risk, as well as differences across injury types and contexts.

The statistical analysis aligned well with the original plan. Hypothesis testing confirmed that both numerical and categorical variables were significantly associated with hospital admission. The t-test showed a significant difference in mean age between admitted and non-admitted patients, while chi-square tests indicated meaningful associations across multiple categorical variables. Logistic regression further demonstrated that age remained the strongest predictor, even after controlling for other factors.

These findings are consistent with existing knowledge in healthcare analytics, where older patients and more severe injury types are more likely to require hospital admission. This reinforces the validity of the dataset and the analytical approach used in this study.

During the project, some deviations from the original proposal occurred. While the initial goal emphasized a simple and interpretable baseline model, the team extended the modeling process to include additional iterations, such as class weighting and feature engineering. This was necessary due to class imbalance in the dataset, where admitted cases were relatively rare. The baseline model performed poorly in identifying admitted patients, which motivated further improvements.

By introducing class weighting and enhanced features (such as interaction terms and contextual variables), the model significantly improved its ability to detect high-risk cases. This shift reflects a practical trade-off between simplicity and performance, highlighting the importance of adapting modeling strategies based on data characteristics.

Another challenge involved balancing statistical rigor with practical usability. While statistical tests provided interpretable insights, they were limited in capturing complex relationships between variables. The integration of machine learning allowed for improved predictive performance, while the deployment of a Streamlit application translated the model into a usable tool.

Despite these successes, several limitations remain. The dataset lacks detailed clinical information, such as patient comorbidities, which could further improve prediction accuracy. Additionally, some subgroup analyses may be influenced by small sample sizes, requiring cautious interpretation.

In summary, the project not only achieved its original analytical goals but also extended beyond them by incorporating predictive modeling and deployment. This demonstrates how statistical analysis and machine learning can complement each other to provide both insight and practical application in healthcare decision-making. A key limitation of this analysis is that NEISS uses a complex stratified probability sampling design. Standard statistical tests applied here assume simple random sampling, and results should therefore be interpreted as approximate rather than strictly inferential.

6. Conclusion

This project demonstrates that hospital admission risk can be effectively predicted using emergency department injury data from the NEISS 2024 dataset. By integrating data cleaning, exploratory analysis, and iterative modeling, the study identified key factors associated with injury severity and patient outcomes.

Age emerged as the strongest predictor of hospital admission, with substantially higher risk observed among older adults. Injury characteristics, including body part and diagnosis, provided additional predictive value, highlighting the clinical importance of injury context.

Through iterative modeling, performance was improved by addressing class imbalance and incorporating contextual and interaction features. The final model achieved strong recall in identifying admitted patients, supporting its usefulness for early detection of high-risk cases, despite the trade-off with precision.

Overall, the results indicate that injury-related factors and age are the primary drivers of admission risk, while demographic variables play a more limited role. The developed model demonstrates practical potential as a decision-support tool, offering a data-driven approach to assist in early triage and resource allocation in emergency care settings.

7. Contributions

Task	Responsibility
Project design and research questions	All group members
Data cleaning and preprocessing	Pei-Ru Chen
Exploratory data analysis	Foluso Ojo
Statistical modeling	Ameenat Ali
Machine learning modeling	Pei-Ru Chen
Visualization and interpretation	Gurpreet Kaur
Clinical decision support tool	Foluso Ojo
Final report	Ameenat Ali
Presentation	All group members

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9. Appendix

The full project code, including data preprocessing, statistical analysis, model development, and deployment, is available at:

https://github.com/pearlnogh/NEISS_2024_Injury_Analysis.git